Numerical Methods for Solving Particle Transport Problems

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Outline

1. Nonlinear methods for solving the particle transport equation
   - The quasidiffusion (QD) method
     - Formulation of the QD method
     - Independent discretization of the QD equations
     - Consistent discretization of the QD equations
   - Nonlinear diffusion acceleration (NDA) method
     - Formulation of the NDA method in continuous form
     - Formulation of the NDA method in discrete form
   - Partial-current-based nonlinear diffusion acceleration (pNDA) method
     - Formulation of the pNDA method
2. Synthetic Acceleration
   - General formulation of Synthetic Acceleration Methods
   - Diffusion Synthetic Acceleration
   - Transport Synthetic Acceleration

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Transport problem for 1D slab geometry

We consider one-group transport equation for 1D slab geometry:

$$\mathcal{L}\psi = \mathcal{S}\psi + Q\,, \tag{1}$$

$$0 \le x \le X\,,$$

$$\mathcal{L}\psi(x,\mu) \equiv \mu\frac{\partial\psi(x,\mu)}{\partial x} + \Sigma_t(x)\psi(x,\mu)\,, \tag{2}$$

$$\mathcal{S}\psi(x,\mu) \equiv \frac{1}{2}\Sigma_s(x)\int_{-1}^{1}\psi(x,\mu')d\mu' \tag{3}$$

$$Q = \frac{1}{2}q \tag{4}$$

$$\psi(0,\mu) = \psi_{in}^{+}(\mu) \text{ for } \mu > 0\,, \quad \psi(X,\mu) = \psi_{in}^{-}(\mu) \text{ for } \mu < 0\,. \tag{5}$$

The scalar flux and current are defined by

$$\phi(x) = \int_{-1}^{1}\psi(x,\mu)d\mu\,, \quad J(x) = \int_{-1}^{1}\mu\psi(x,\mu)d\mu\,.$$

Algorithm of source iterations

$s = -1$

Set $\phi^{(0)} = \phi^{initial}$

while $||\phi^{(s)} - \phi^{(s-1)}||_\infty > \varepsilon_{\phi,1}||\phi^{(s)}||_\infty + \varepsilon_{\phi,2}$ **do**

$s = s + 1$

1. Solve the transport problem

$$\mathcal{L}\psi^{(s+1/2)} = \frac{1}{2}\left(\Sigma_s \phi^{(s)} + q\right) \ ,$$

$$\psi^{(s+1/2)}(0,\mu) = \psi^{+}_{in}(\mu)\,, \text{ for } \mu > 0\,, \quad \psi^{(s+1/2)}(X,\mu) = \psi^{-}_{in}(\mu)\,, \text{ for } \mu < 0\,.$$

to calculate $\psi^{(s+1/2)}$ using $\phi^{(s)}$ to compute its RHS

2. Compute $\phi^{(s+1/2)} = \int_{-1}^{1} \psi^{(s+1/2)} d\mu$

3. Set $\phi^{(s+1)} = \phi^{(s+1/2)}$

end

QD method and exact closures

The Quasidiffusion (QD) method is a nonlinear method with exact closures for solving particle transport problems defined by the following system of equations (V. Gol'din, 1964; L. Auer & D. Mihalas, 1970)

- the high-order transport equation

$$\mathcal{L}\psi = \frac{1}{2}\left(\Sigma_s\phi + q\right) \,, \tag{6}$$

- the low-order QD (LOQD) equations (QD moment equations)

$$\frac{d}{dx}J(x) + (\Sigma_t - \Sigma_s)\phi(x) = q(x) \,, \tag{7}$$

$$\frac{d}{dx}\Big(E(x)\phi(x)\Big) + \Sigma_t J(x) = 0 \,, \tag{8}$$

- BCs for the LOQD equations

$$J(0) = C_L(\phi(0) - \phi^+_{in}) + J^+_{in} \,, \quad J(X) = C_R(\phi(X) - \phi^-_{in}) + J^-_{in} \,, \tag{9}$$

- the QD (Eddington) and boundary factors

$$E(x) = \frac{\int_{-1}^{1}\mu^2\psi(x,\mu)d\mu}{\int_{-1}^{1}\psi(x,\mu)d\mu} \qquad C_L = \frac{\int_{-1}^{0}\mu\psi(0,\mu)d\mu}{\int_{-1}^{0}\psi(0,\mu)d\mu} \,, \quad C_R = \frac{\int_{0}^{1}\mu\psi(X,\mu)d\mu}{\int_{0}^{1}\psi(X,\mu)d\mu} \,, \tag{10}$$

- partial fluxes and currents of incoming particles

$$\phi^\pm_{in} = \pm\int_0^{\pm1}\psi^\pm_{in}(\mu)d\mu \,, \quad J^\pm_{in} = \int_0^{\pm1}\mu\psi^\pm_{in}(\mu)d\mu \,. \tag{11}$$

QD iteration algorithm

1. A transport sweep

$$\mathcal{L}\psi^{(s+1/2)} = \frac{1}{2}\left(\Sigma_s \phi^{(s)} + q\right) \,, \tag{12}$$

$$\psi^{(s+1/2)}(0,\mu) = \psi^{+}_{in}(\mu)\,, \text{ for } \mu > 0\,, \quad \psi^{(s+1/2)}(X,\mu) = \psi^{-}_{in}(\mu)\,, \text{ for } \mu < 0\,. \tag{13}$$

2. Calculation of the QD factors

$$E^{(s+1/2)}(x) = \frac{\int_{-1}^{1} \mu^2 \psi^{(s+1/2)}(x,\mu)d\mu}{\int_{-1}^{1} \psi^{(s+1/2)}(x,\mu)d\mu} \,, \tag{14}$$

$$C_L^{(s+1/2)} = \frac{\int_{-1}^{0} \mu\psi^{(s+1/2)}(0,\mu)d\mu}{\int_{-1}^{0} \psi^{(s+1/2)}(0,\mu)d\mu} \,, \quad C_R^{(s+1/2)} = \frac{\int_{0}^{1} \mu\psi^{(s+1/2)}(X,\mu)d\mu}{\int_{0}^{1} \psi^{(s+1/2)}(X,\mu)d\mu} \,. \tag{15}$$

3. Solving the LOQD problem for $\phi^{(s+1)}$ and $J^{(s+1)}$

$$\frac{d}{dx}J^{(s+1)} + (\Sigma_t - \Sigma_s)\phi^{(s+1)} = q \,, \tag{16}$$

$$\frac{d}{dx}\left(E^{(s+1/2)}\phi^{(s+1)}\right) + \Sigma_t J^{(s+1)} = 0 \,, \tag{17}$$

$$J^{(s+1)}(0) = C_L^{(s+1/2)}(\phi^{(s+1)}(0) - \phi^{+}_{in}) + J^{+}_{in} \,, \tag{18}$$

$$J^{(s+1)}(X) = C_R^{(s+1/2)}(\phi^{(s+1)}(X) - \phi^{-}_{in}) + J^{-}_{in} \,. \tag{19}$$

QD iteration algorithm

$s = -1$

Set $\phi^{(0)} = \phi^{initial}$

while $||\phi^{(s)} - \phi^{(s-1)}||_\infty > \varepsilon_{\phi,1}||\phi^{(s)}||_\infty + \varepsilon_{\phi,2}$ **do**

$s = s + 1$

1. Solve the high-order transport problem to calculate $\psi^{(s+1/2)}$ using $\phi^{(s)}$ to compute its RHS
2. Compute $E^{(s+1/2)}$, $C_R^{(s+1/2)}$ and $C_L^{(s+1/2)}$ using $\psi^{(s+1/2)}$
3. Solve the low-order QD problem for $\phi^{(s+1)}$ and $J^{(s+1)}$ using $E^{(s+1/2)}$, $C_L^{(s+1/2)}$ and $C_R^{(s+1/2)}$ as coefficients of the LOQD equations

end

Independent discretization of the QD equations

- We define
  - a spatial mesh: $\{x_{i+1/2}, i = 1, ..., N, x_{1/2} = 0, x_{N+1/2} = X\}$
  - an angular quadrature set: $\{\mu_m, w_m, m = 1, ..., M\}$.

- The high-order transport equation (6) is approximated by means of the step characteristic (SC) method

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi_i + q_i)\Delta x_i \,, \tag{20}$$

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1 - \alpha_{m,i})\psi_{m,i+1/2} \,, \tag{21}$$

$$\alpha_{m,i} = \frac{1}{\tau_{m,i}} - \frac{1}{e^{\tau_{m,i}} - 1} \,, \tag{22}$$

$$\tau_{m,i} = \frac{\Sigma_{t,i}\Delta x_i}{\mu_m} \,, \tag{23}$$

$$\Delta x_i = x_{i+1/2} - x_{i-1/2} \,, \tag{24}$$

$$i = 1, ..., N \,, \quad m = 1, ..., M \,,$$

$$\psi_{m,1/2} = \psi^+_{in,m} \,, \;\; m : \;\; \mu_m > 0 \,, \tag{25}$$

$$\psi_{m,N+1/2} = \psi^-_{in,m} \,, \;\; m : \;\; \mu_m < 0 \,. \tag{26}$$

Independent discretization of the QD equations

The cell-average and cell-edge QD factors are approximated as follows:

$$E_i = \frac{\sum_{m=1}^{M} \mu_m^2 \psi_{m,i} w_m}{\sum_{m=1}^{M} \psi_{m,i} w_m}, \quad i = 1, ..., N, \tag{27}$$

$$E_{i+1/2} = \frac{\sum_{m=1}^{M} \mu_m^2 \psi_{m,i+1/2} w_m}{\sum_{m=1}^{M} \psi_{m,i+1/2} w_m}, \quad i = 0, ..., N. \tag{28}$$

The boundary QD factors are given by

$$C_L = \frac{\sum_{\{m:\mu_m \leq 0\}} \mu_m \psi_{m,1/2} w_m}{\sum_{\{m:\mu_m \leq 0\}} \psi_{m,1/2} w_m}, \tag{29}$$

$$C_R = \frac{\sum_{\{m:\mu_m \geq 0\}} \mu_m \psi_{m,N+1/2} w_m}{\sum_{\{m:\mu_m \geq 0\}} \psi_{m,N+1/2} w_m}. \tag{30}$$

Independent discretization of the QD equations

• The LOQD equations (7)-(8) are discretized by a second-order finite volume (FV) scheme.

• The balance equation (7) is integrated over ith spatial cell ($x_{i-1/2} \leq x \leq x_{i+1/2}$) to get

$$J_{i+1/2} - J_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi_i = q_i \Delta x_i \,. \tag{31}$$

• We now integrate the first moment equation (8) over the left half of the ith cell ($x_{i-1/2} \leq x \leq x_i$) to obtain

$$E_i \phi_i - E_{i-1/2}\phi_{i-1/2} + \frac{1}{2}\Sigma_{t,i}\Delta x_i J_{i-1/2} = 0\,, \tag{32}$$

where

$$x_i = 0.5(x_{i-1/2} + x_{i+1/2})\,. \tag{33}$$

• Then we integrate (8) over the right half of the ith cell ($x_i \leq x \leq x_{i+1/2}$) to get

$$E_{i+1/2}\phi_{i+1/2} - E_i \phi_i + \frac{1}{2}\Sigma_{t,i}\Delta x_i J_{i+1/2} = 0\,. \tag{34}$$

• The boundary conditions have the following form:

$$J_{1/2} = C_L(\phi_{1/2} - \phi_{in}^+) + J_{in}^+\,, \quad J_{N+1/2} = C_R(\phi_{N+1/2} - \phi_{in}^-) + J_{in}^-\,. \tag{35}$$

Independent discretization of the QD equations

- The system of discretized low-order QD equations are given by
  - the balance equation

$$J_{i+1/2} - J_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi_i = q_i \Delta x_i \,. \tag{36}$$

  - the first-moment equations

$$E_i \phi_i - E_{i-1/2}\phi_{i-1/2} + \frac{1}{2}\Sigma_{t,i}\Delta x_i J_{i-1/2} = 0\,, \tag{37}$$

$$E_{i+1/2}\phi_{i+1/2} - E_i \phi_i + \frac{1}{2}\Sigma_{t,i}\Delta x_i J_{i+1/2} = 0\,. \tag{38}$$

  - LOQD boundary conditions

$$J_{1/2} = C_L(\phi_{1/2} - \phi_{in}^+) + J_{in}^+\,, \quad J_{N+1/2} = C_R(\phi_{N+1/2} - \phi_{in}^-) + J_{in}^-\,. \tag{39}$$

- The discretized high-order transport equation

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi_i + q_i)\Delta x_i\,, \tag{40}$$

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1 - \alpha_{m,i})\psi_{m,i+1/2}\,, \tag{41}$$

Idea of consistent discretization

- Continuous equations
  - high-order equation

$$\mathcal{L}\psi = \frac{1}{2}\left(\Sigma_s\phi + q\right) \ , \tag{42}$$

  - projection operators (integration) for equations

$$\int_{-1}^{1} \cdot \, d\mu \, , \quad \int_{-1}^{1} \cdot \, \mu d\mu \, , \tag{43}$$

  - projection operators for the low-order solutions (moments)

$$\phi = \int_{-1}^{1} \psi d\mu \, , \quad J = \int_{-1}^{1} \mu\psi d\mu \, , \tag{44}$$

  - low-order equations

$$\frac{d}{dx}J(x) + (\Sigma_t - \Sigma_s)\phi(x) = q(x) \ , \tag{45}$$

$$\frac{d}{dx}\Big(E(x)\phi(x)\Big) + \Sigma_t J(x) = 0 \ , \tag{46}$$

  - closures

$$E(x) = \frac{\int_{-1}^{1} \mu^2\psi(x,\mu)d\mu}{\int_{-1}^{1} \psi(x,\mu)d\mu} \ . \tag{47}$$

Idea of consistent discretization

- Discretization starts from high-order equation

$$\mathcal{L}_h \psi_h = \frac{1}{2} \left(\Sigma_s \phi_h + q_h \right) \ . \tag{48}$$

- The approximation of the low-order equations is driven by the transport discretization scheme for the high-order equation
- Projection of the transport equation in discrete space is an approximation of the projection in continuous space

$$\int_{-1}^{1} \cdot \, d\mu \Longrightarrow \sum_m \cdot \, w_m \qquad \int_{-1}^{1} \cdot \, \mu d\mu \Longrightarrow \sum_m \cdot \, \mu_m w_m \, . \tag{49}$$

- Projection operator for the solution of the low-order equations

$$\phi_h = \sum_m \psi_h w_m \, , \quad J_h = \sum_m \mu_m \psi_h w_m \, . \tag{50}$$

- The discretization of the balance equation is defined by

$$\sum_m \mathcal{L}_h \psi_h w_m = \frac{1}{2} \sum_m \left(\Sigma_s \phi_h + q_h \right) w_m \ . \tag{51}$$

- The discretization of the first moment equations is based on

$$\sum_m \mu_m \mathcal{L}_h \psi_h w_m = \frac{1}{2} \sum_m \mu_m \left(\Sigma_s \phi_h + q_h \right) w_m \ . \tag{52}$$

- Auxiliary equations for discretized low-order result from projection of the auxiliary equations of transport discretization scheme
- Closures are formulated directly for the projected discretized equations

Consistent discretization of the QD equations

- We now discretize the LOQD equations in space by a scheme which is *algebraically consistent* with the SC transport method.
- To derive such a consistent scheme we apply to the given transport scheme operators that are discrete analog of those used to derive the LOQD equation from the continuous transport equation.
- First we integrate the discretized transport equation (20) in the discrete space

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi_i + q_i)\Delta x_i \,, \tag{53}$$

over angular variable by applying discrete operator $\sum_{m=1}^{M} \cdot w_m$ to get

$$J_{i+1/2} - J_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi_i = q_i \Delta x_i \,, \tag{54}$$

where

$$J_{i+1/2} = \sum_{m=1}^{M} \mu_m \psi_{m,i+1/2} w_m \,, \tag{55}$$

$$\phi_i = \sum_{m=1}^{M} \psi_{m,i} w_m \,. \tag{56}$$

Consistent discretization of the QD equations

- Secondly we integrate Eq. (20)

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi_i + q_i)\Delta x_i \,, \tag{57}$$

with weight μ_m and hence apply discrete operator $\sum_{m=1}^{M} \cdot \mu_m w_m$ to obtain

$$\sum_{m=1}^{M} \mu_m^2 \psi_{m,i+1/2} w_m - \sum_{m=1}^{M} \mu_m^2 \psi_{m,i+1/2} w_m + \Sigma_{t,i}\Delta x_i J_i = 0 \,, \tag{58}$$

where

$$J_i = \sum_{m=1}^{M} \mu_m \psi_{m,i} w_m \,. \tag{59}$$

- We use the cell-edge QD factors

$$E_{i+1/2} = \frac{\sum_{m=1}^{M} \mu_m^2 \psi_{m,i+1/2} w_m}{\sum_{m=1}^{M} \psi_{m,i+1/2} w_m} \tag{60}$$

to get

$$E_{i+1/2}\phi_{i+1/2} - E_{i-1/2}\phi_{i-1/2} + \Sigma_{t,i}\Delta x_i J_i = 0 \,, \tag{61}$$

$$\phi_{i+1/2} = \sum_{m=1}^{M} \psi_{m,i+1/2} w_m \,. \tag{62}$$

Consistent discretization of the QD equations

- To formulate auxiliary relations for those unknowns we integrate

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1-\alpha_{m,i})\psi_{m,i+1/2}\,, \tag{63}$$

with weight μ_m and μ_m^2 to get

$$J_i = \sum_{m=1}^{M} \mu_m \alpha_{m,i}\psi_{m,i-1/2} w_m + \sum_{m=1}^{M} \mu_m (1-\alpha_{m,i})\psi_{m,i+1/2} w_m\,, \tag{64}$$

$$\sum_{m=1}^{M} \mu_m^2 \psi_{m,i} w_m = \sum_{m=1}^{M} \mu_m^2 \alpha_{m,i}\psi_{m,i-1/2} w_m + \sum_{m=1}^{M} \mu_m^2 (1-\alpha_{m,i})\psi_{m,i+1/2} w_m\,. \tag{65}$$

- We define the following auxiliary factors

$$F_i^- = \frac{\sum_{m=1}^{M} \mu_m \alpha_{m,i}\psi_{m,i-1/2} w_m}{\sum_{m=1}^{M} \psi_{m,i-1/2} w_m}\,, \quad F_i^+ = \frac{\sum_{m=1}^{M} \mu_m (1-\alpha_{m,i})\psi_{m,i+1/2} w_m}{\sum_{m=1}^{M} \psi_{m,i+1/2} w_m}\,, \tag{66}$$

$$A_i^- = \frac{\sum_{m=1}^{M} \mu_m^2 \alpha_{m,i}\psi_{m,i-1/2} w_m}{\sum_{m=1}^{M} \mu_m^2 \psi_{m,i-1/2} w_m}\,, \quad A_i^+ = \frac{\sum_{m=1}^{M} \mu_m^2 (1-\alpha_{m,i})\psi_{m,i+1/2} w_m}{\sum_{m=1}^{M} \mu_m^2 \psi_{m,i+1/2} w_m}\,. \tag{67}$$

- The auxiliary relations for LOQD equations

$$J_i = F_i^- \phi_{i-1/2} + F_i^+ \phi_{i+1/2}\,, \tag{68}$$

$$E_i\phi_i = A_i^- E_{i-1/2}\phi_{i-1/2} + A_i^+ E_{i+1/2}\phi_{i+1/2}\,. \tag{69}$$

Consistent discretization of the QD equations

- The resulting consistently discretized LOQD equations are given by
  - the balance equation

$$J_{i+1/2} - J_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi_i = q_i \Delta x_i \,. \tag{70}$$

  - the first-moment equation

$$E_{i+1/2}\phi_{i+1/2} - E_{i-1/2}\phi_{i-1/2} + \Sigma_{t,i}\Delta x_i J_i = 0 \,, \tag{71}$$

  - auxiliary relations

$$J_i = F_i^- \phi_{i-1/2} + F_i^+ \phi_{i+1/2} \,, \tag{72}$$

$$E_i \phi_i = A_i^- E_{i-1/2}\phi_{i-1/2} + A_i^+ E_{i+1/2}\phi_{i+1/2} \,, \tag{73}$$

  - LOQD boundary conditions

$$J_{1/2} = C_L(\phi_{1/2} - \phi_{in}^+) + J_{in}^+ \,, \quad J_{N+1/2} = C_R(\phi_{N+1/2} - \phi_{in}^-) + J_{in}^- \,. \tag{74}$$

- The discretized high-order transport equation

$$\mu_m(\psi_{m,i+1/2} - \psi_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi_i + q_i)\Delta x_i \,, \tag{75}$$

$$\psi_{m,i} = \alpha_{m,i}\psi_{m,i-1/2} + (1 - \alpha_{m,i})\psi_{m,i+1/2} \,. \tag{76}$$

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Formulation of the NDA Method in continuous form

- High-order problem

$$\mathcal{L}\psi = \frac{1}{2}(\Sigma_s\phi + q)\,, \tag{77}$$

- Low-order problem

$$\frac{dJ}{dx} + (\Sigma_t - \Sigma_s)\,\phi = q\,, \tag{78}$$

$$J = -\frac{1}{3\Sigma_t}\frac{d\phi}{dx} + \tilde{D}\phi\,, \tag{79}$$

where the closure for the compensation term is defined by

$$\tilde{D} = \frac{1}{\int_{-1}^{1}\psi d\mu}\left(\int_{-1}^{1}\mu\psi d\mu + \frac{1}{3\Sigma_t}\frac{d}{dx}\left[\int_{-1}^{1}\psi d\mu\right]\right) \tag{80}$$

(compensation factor).

- The idea of the NDA is to formulate an equation based on a Fick's law relation with an exact closure

$$\frac{1}{3\Sigma_t}\frac{d\phi}{dx} + J = \frac{1}{3\Sigma_t}\frac{d\tilde{\phi}}{dx} + \tilde{J}\,, \quad \tilde{\phi} \equiv \int_{-1}^{1}\psi d\mu\,, \quad \tilde{J} \equiv \int_{-1}^{1}\mu\psi d\mu\,, \tag{81}$$

$$\Longrightarrow \quad \frac{1}{3\Sigma_t}\frac{d\phi}{dx} + J = \frac{1}{\tilde{\phi}}\left(\frac{1}{3\Sigma_t}\frac{d\tilde{\phi}}{dx} + \tilde{J}\right)\phi \qquad \tilde{D} = \frac{1}{\tilde{\phi}}\left(\tilde{J} + \frac{1}{3\Sigma_t}\frac{d\tilde{\phi}}{dx}\right)\,. \tag{82}$$

Formulation of the NDA method in continuous form

$$\mathcal{L}\psi^{(s+1/2)} = \frac{1}{2}(\Sigma_s \phi^{(s)} + q)\,, \tag{83}$$

$$\phi^{(s+1/2)} = \int_{-1}^{1} \psi^{(s+1/2)} d\mu\,, \quad J^{(s+1/2)} = \int_{-1}^{1} \mu\psi^{(s+1/2)} d\mu\,, \tag{84}$$

$$\tilde{D}^{(s+1/2)} = \frac{1}{\phi^{(s+1/2)}} \left(J^{(s+1/2)} + \frac{1}{3\Sigma_t} \frac{d\phi^{(s+1/2)}}{dx} \right)\,, \tag{85}$$

$$\frac{dJ^{(s+1)}}{dx} + (\Sigma_t - \Sigma_s)\,\phi^{(s+1)} = q\,, \tag{86}$$

$$J^{(s+1)} = -\frac{1}{3\Sigma_t} \frac{d\phi^{(s+1)}}{dx} + \tilde{D}^{(s+1/2)} \phi^{(s+1)}\,, \tag{87}$$

Formulation of the NDA method in discrete form

The NDA method is defined by the following system of discrete equations (K. Smith, 1983; K. Smith and J. Rhodes, 2002):

- the discrete high-order transport problem

$$\mu_m(\psi^{(s+1/2)}_{m,i+1/2} - \psi^{(s+1/2)}_{m,i-1/2}) + \Sigma_{t,i}\Delta x_i \psi^{(s+1/2)}_{m,i} = \frac{1}{2}(\Sigma_{s,i}\phi^{(s)}_i + q_i)\Delta x_i \,, \tag{88}$$

$$\psi^{(s+1/2)}_{m,i} = \alpha_{m,i}\psi^{(s+1/2)}_{m,i-1/2} + (1-\alpha_{m,i})\psi^{(s+1/2)}_{m,i+1/2} \,, \tag{89}$$

$$\alpha_{m,i} = \frac{1}{\tau_{m,i}} - \frac{1}{e^{\tau_{m,i}} - 1} \,, \tag{90}$$

- the discrete low-order NDA (LONDA) problem

$$J^{(s+1)}_{i+1/2} - J^{(s+1)}_{i-1/2} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi^{(s+1)}_i = q_i \Delta x_i \,, \tag{91}$$

$$J^{(s+1)}_{i+1/2} = -D_{i+1/2}\frac{\phi^{(s+1)}_{i+1} - \phi^{(s+1)}_i}{\Delta x_{i+1/2}} + \tilde{D}^{(s+1/2)}_{i+1/2}(\phi^{(s+1)}_{i+1} + \phi^{(s+1)}_i) \,. \tag{92}$$

Formulation of the NDA method in discrete form

- The factor $\tilde{D}_{i+1/2}$ is chosen to satisfy

$$\tilde{J}_{i+1/2} = -D_{i+1/2}\frac{\tilde{\phi}_{i+1} - \tilde{\phi}_i}{\Delta x_{i+1/2}} + \tilde{D}_{i+1/2}(\tilde{\phi}_{i+1} + \tilde{\phi}_i)\,, \tag{93}$$

where

$$\tilde{J}_{i+1/2} = \sum_m \mu_m \psi_{m,i+1/2} w_m\,, \quad \tilde{\phi}_i = \sum_m \psi_{m,i} w_m \tag{94}$$

are the high-order moments,

$$D_{i+1/2} = \frac{1}{3\Sigma_{t,i+1/2}}\,, \quad \Sigma_{t,i+1/2} = \frac{\Sigma_{t,i}\Delta x_i + \Sigma_{t,i+1}\Delta x_{i+1}}{\Delta x_i + \Delta x_{i+1}}\,, \tag{95}$$

$$\Delta x_{i+1/2} = \frac{1}{2}(\Delta x_i + \Delta x_{i+1})\,, \quad \Delta x_0 = 0\,, \quad \Delta x_{N+1} = 0\,. \tag{96}$$

- The NDA factor is defined as

$$\tilde{D}^{(s+1/2)}_{i+1/2} = \frac{1}{\phi^{(s+1/2)}_{i+1} + \phi^{(s+1/2)}_i}\left(J^{(s+1/2)}_{i+1/2} + D_{i+1/2}\frac{(\phi^{(s+1/2)}_{i+1} - \phi^{(s+1/2)}_i)}{\Delta x_{i+1/2}}\right)\,. \tag{97}$$

$$J^{(s+1/2)}_{i+1/2} = \sum_m \mu_m \psi^{(s+1/2)}_{m,i+1/2} w_m\,, \quad \phi^{(s+1/2)}_i = \sum_m \psi^{(s+1/2)}_{m,i} w_m\,. \tag{98}$$

NDA iteration algorithm

$s = -1$

Set $\phi^{(0)} = \phi^{initial}$

1 **while** $||\phi^{(s)} - \phi^{(s-1)}||_\infty > \varepsilon_{\phi,1}||\phi^{(s)}||_\infty + \varepsilon_{\phi,2}$ **do**

$s = s + 1$

1. Solve the high-order transport problem to calculate $\psi^{(s+1/2)}$ using $\phi^{(s)}$ to compute its RHS

2. Compute $\tilde{D}^{(s+1/2)}$, $C_R^{(s+1/2)}$ and $C_L^{(s+1/2)}$ using $\psi^{(s+1/2)}$

3. Solve the low-order **NDA** problem for $\phi^{(s+1)}$ and $J^{(s+1)}$ using $\tilde{D}^{(s+1/2)}$, $C_L^{(s+1/2)}$ and $C_R^{(s+1/2)}$ as coefficients of the LONDA equations

end

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Formulation of the pNDA method in discrete form

There exists another version of the NDA method which uses partial currents to derive the LONDA equations (N. Z. Cho, G. S. Lee, and C. J. Park, 2003.)

The angular flux in general P_1 approximation

$$\psi(x,\mu) = \frac{1}{2}\phi(x) + \frac{3}{2}\mu J(x)\,. \tag{99}$$

The partial currents

$$J^- = \int_{-1}^{0} \mu\psi d\mu\,, \quad J^+ = \int_{0}^{1} \mu\psi d\mu\,, \tag{100}$$

$$J = J^- + J^+\,.$$

The partial currents in the P_1 approximation

$$J^- = -\frac{1}{4}\phi + \frac{1}{2}J\,, \quad J^+ = \frac{1}{4}\phi + \frac{1}{2}J\,. \tag{101}$$

Taking into account the Fick's law we obtain

$$J^- = -\frac{1}{2}D\frac{d\phi}{dx} - \frac{1}{4}\phi\,, \quad J^+ = -\frac{1}{2}D\frac{d\phi}{dx} + \frac{1}{4}\phi\,. \tag{102}$$

Formulation of the pNDA method in discrete form

The pNDA factors $\tilde{D}^{\pm}_{i+1/2}$ are defined to meet the following equations:

$$\tilde{J}^{-}_{i+1/2} = -\frac{D_{i+1/2}}{2\Delta x_{i+1/2}}\left(\tilde{\phi}_{i+1} - \tilde{\phi}_i\right) - \tilde{D}^{-}_{i+1/2}\tilde{\phi}_{i+1}\,, \tag{103}$$

$$\tilde{J}^{+}_{i+1/2} = -\frac{D_{i+1/2}}{2\Delta x_{i+1/2}}\left(\tilde{\phi}_{i+1} - \tilde{\phi}_i\right) + \tilde{D}^{+}_{i+1/2}\tilde{\phi}_i\,. \tag{104}$$

Note that

$$J_{i+1/2} = \tilde{J}^{+}_{i+1/2} + \tilde{J}^{-}_{i+1/2} = -D_{i+1/2}\frac{\phi_{i+1} - \phi_i}{\Delta x_{i+1/2}} - \tilde{D}^{-}_{i+1/2}\phi_{i+1} + \tilde{D}^{+}_{i+1/2}\phi_i\,. \tag{105}$$

The pNDA factors are defined as

$$\tilde{D}^{-(s+1/2)}_{i+1/2} = -\frac{1}{\tilde{\phi}^{(s+1/2)}_{i+1}}\left(\tilde{J}^{-(s+1/2)}_{i+1/2} + 0.5D_{i+1/2}\frac{(\tilde{\phi}^{(s+1/2)}_{i+1} - \tilde{\phi}^{(s+1/2)}_{i})}{\Delta x_{i+1/2}}\right)\,, \tag{106a}$$

$$\tilde{D}^{+(s+1/2)}_{i+1/2} = \frac{1}{\tilde{\phi}^{(s+1/2)}_{i+1}}\left(\tilde{J}^{+(s+1/2)}_{i+1/2} + 0.5D_{i+1/2}\frac{(\tilde{\phi}^{(s+1/2)}_{i+1} - \tilde{\phi}^{(s+1/2)}_{i})}{\Delta x_{i+1/2}}\right)\,. \tag{106b}$$

Formulation of the pNDA method in discrete form

The discrete low-order pNDA problem

$$J_{i+1/2}^{(s+1)} - J_{i-1/2}^{(s+1)} + (\Sigma_{t,i} - \Sigma_{s,i})\Delta x_i \phi_i^{(s+1)} = q_i \Delta x_i \,, \tag{107a}$$

$$i = 1, ..., N \,,$$

$$J_{i+1/2}^{(s+1)} = -D_{i+1/2} \frac{\phi_{i+1}^{(s+1)} - \phi_i^{(s+1)}}{\Delta x_{i+1/2}} - \tilde{D}_{i+1/2}^{-(s+1/2)} \phi_{i+1}^{(s+1)} + \tilde{D}_{i+1/2}^{+(s+1/2)} \phi_i^{(s+1)} \,, \tag{107b}$$

$$i = 0, ..., N \,.$$

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Synthetic Acceleration Methods

(J. Kopp, 1963)

$$\boxed{\mathcal{L}\psi = \mathcal{S}\psi + Q} \tag{108}$$

Source iterations

$$\mathcal{L}\psi^{(s+1/2)} = \mathcal{S}\psi^{(s)} + Q\,, \quad \psi^{(s+1)} = \psi^{(s+1/2)} \tag{109}$$

Source iterations for iterative errors

$$\delta\psi^{(s+1/2)} = \psi - \psi^{(s+1/2)} \tag{110}$$

$$\mathcal{L}\psi - \mathcal{L}\psi^{(s+1/2)} = \mathcal{S}\psi + Q - \mathcal{S}\psi^{(s)} - Q \tag{111}$$

$$\mathcal{L}\delta\psi^{(s+1/2)} = \mathcal{S}\delta\psi^{(s+1/2)} + \mathcal{S}\left(\psi^{(s+1/2)} - \psi^{(s)}\right) \tag{112}$$

$$\boxed{\delta\psi^{(s+1/2)} = \left(\mathcal{L} - \mathcal{S}\right)^{-1}\mathcal{S}\left(\psi^{(s+1/2)} - \psi^{(s)}\right)} \tag{113}$$

$$\psi^{(s+1)} = \psi^{(s+1/2)} + \delta\psi^{(s+1/2)} \tag{114}$$

Synthetic Acceleration (SA) methods for iterative correction f

$$\mathcal{M} \approx \left(\mathcal{L} - \mathcal{S}\right)^{-1} \tag{115}$$

$$\boxed{f^{(s+1/2)} = \mathcal{M}\mathcal{S}\left(\psi^{(s+1/2)} - \psi^{(s)}\right)} \tag{116}$$

$$\psi^{(s+1)} = \psi^{(s+1/2)} + f^{(s+1/2)} \tag{117}$$

$$\psi^{(s+1)} = \psi^{(s+1/2)} + \mathcal{M}\mathcal{S}\left(\psi^{(s+1/2)} - \psi^{(s)}\right) \tag{118}$$

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Diffusion Synthetic Acceleration (DSA)

(J. Kopp, 1963; V. Lebedev, 1964; R. Alcouffe, 1976)

$$\mathcal{L}\psi^{(s+1/2)} = \mathcal{S}\psi^{(s)} + Q \tag{119}$$

$$\mathcal{L}\delta\psi^{(s+1/2)} = \mathcal{S}\delta\psi^{(s+1/2)} + \mathcal{S}\left(\psi^{(s+1/2)} - \psi^{(s)}\right) \tag{120}$$

$$\mu\frac{\partial\delta\psi^{(s+1/2)}}{\partial x} + \Sigma_t\delta\psi^{(s+1/2)} = \frac{1}{2}\Sigma_s\int_{-1}^{1}\delta\psi^{(s+1/2)}(x,\mu')d\mu' + \frac{1}{2}\Sigma_s\left(\phi^{(s+1/2)} - \phi^{(s)}\right) \tag{121}$$

The basic equation for the iterative correction $f(x,\mu)$

$$\mu\frac{\partial f^{(s+1/2)}}{\partial x}(x,\mu) + \Sigma_t f^{(s+1/2)}(x,\mu) = \frac{1}{2}\Sigma_s\int_{-1}^{1} f^{(s+1/2)}(x,\mu')d\mu' + \frac{1}{2}\Sigma_s\left(\phi^{(s+1/2)} - \phi^{(s)}\right) \tag{122}$$

The low-order DSA (LODSA) equation for the iterative correction

$$\boxed{-\frac{d}{dx}\frac{1}{3\Sigma_t}\frac{dF^{(s+1/2)}}{dx}(x) + \Sigma_a F^{(s+1/2)}(x) = \Sigma_s\left(\phi^{(s+1/2)}(x) - \phi^{(s)}(x)\right)} \tag{123}$$

The additive iterative correction for the scalar flux

$$\phi^{(s+1)}(x) = \phi^{(s+1/2)}(x) + F^{(s+1/2)}(x) \tag{124}$$

Diffusion Synthetic Acceleration

High-order transport equation

$$\mathcal{L}\psi^{(s+1/2)} = \frac{1}{2}\left(\Sigma_s \psi^{(s)} + q\right) \tag{125}$$

$$\phi^{(s+1/2)} = \int_{-1}^{1} \psi^{(s+1/2)} d\mu \tag{126}$$

The LODSA equation for the iterative correction

$$-\frac{d}{dx}\frac{1}{3\Sigma_t}\frac{dF^{(s+1/2)}(x)}{dx} + \Sigma_a F^{(s+1/2)}(x) = \Sigma_s\left(\phi^{(s+1/2)} - \phi^{(s)}\right) \tag{127}$$

$$\phi^{(s+1)} = \phi^{(s+1/2)} + F^{(s+1/2)} \tag{128}$$

For stability of iterations, the discretization of the LODSA equations should be **algebraically consistent** with the scheme for the high-order transport equation.

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Transport Synthetic Acceleration (TSA) Method

(V. Morozov, 1962; G. Ramone, M. Adams, P. Nowak, 1997)

High-order transport equation

$$\mathcal{L}\psi^{(s+1/2)} = \mathcal{S}\psi^{(s)} + Q \tag{129}$$

$$\phi^{(s+1/2)} = \int_{-1}^{1} \psi^{(s+1/2)} d\mu \tag{130}$$

The TSA equation for the iterative correction

$$\mu \frac{\partial f_{TSA}^{(s+1/2)}}{\partial x} + (\Sigma_t - \gamma\Sigma_s) f_{TSA}^{(s+1/2)} = \frac{1}{2}(1-\gamma)\Sigma_s \int_{-1}^{1} f_{TSA}^{(s+1/2)}(x,\mu')d\mu' + \frac{1}{2}\Sigma_s\left(\phi^{(s+1/2)} - \phi^{(s)}\right) \tag{131}$$

γ is a parameter

$$\phi^{(s+1)}(x) = \phi^{(s+1/2)}(x) + \int_{-1}^{1} f_{TSA}^{(s+1/2)}(x,\mu)d\mu \tag{132}$$